

Merabi Khachidze

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Portfolio [MerabiKhachidze.github.io](#)

EDUCATION

San Diego State University - San Diego, CA

B.S. Electrical Engineering | May 2026

Relevant Coursework: Electromagnetics, RF & Microwave Engineering, Analog Circuit Design, Circuit Analysis I & II, Digital Systems, Embedded Systems Programming, Fundamentals of Engineering Electronics, Design of Electronic Circuits, Energy & Power Systems, Linear Algebra, Signal Processing

TECHNICAL SKILLS

RF & Analog / Mixed-Signal Design: ADC/DAC architecture & design, noise analysis, impedance matching, op-amp circuits, current to voltage conversion, DAC biasing, PCB layout for analog/RF, flyback transformer design, wireless power transfer (inductive coupling), resonant circuits, power amplifier topologies

EDA & Simulation Tools: Cadence Virtuoso (schematic & layout), LTSpice, KiCad, Altium Designer, MATLAB

HDL & Programming: Verilog (RTL design & simulation), Python, C, C++, MIPS/x86 Assembly

Test & Measurement: Oscilloscope, Waveform Generator, Multimeter, Soldering, PCB bring-up & debug

Hardware Platforms: ATMEL microcontrollers, PIC MCU family, embedded systems integration

PROJECTS

Mixed-Signal Glucose Meter IC Design | Team Lead - ADC/DAC Design | San Diego State University | May 2026

- Designing a fully integrated glucose meter system on-chip in Cadence Virtuoso, targeting a mixed-signal architecture based on the Microchip PIC16LF178X glucose meter reference design
- Developed clean and noisy biosensor signal models in MATLAB, simulating electrochemical current output from the biochemical test strip to define ADC input specifications (dynamic range, noise floor, bandwidth)
- Implemented RTL models of the signal chain in Verilog, including DAC bias voltage generation (VBIAS) and ADC front-end, to verify digital integration before Cadence implementation
- Personally responsible for full custom ADC and DAC design in Cadence - including op-amp-based current-to-voltage converter, RC filter interface to ADC, and DAC reference voltage architecture
- System integrates temperature compensation (analog sensor front-end), EEPROM calibration, and LCD output - providing hands on experience across the full mixed-signal SoC design flow

Switching Flyback Power Supply for Linear Actuator | Student Hardware Lead | San Diego State University / General Atomics

- Designed a switching flyback power supply circuit to meet specific voltage/current requirements for a linear actuator load
- Engineered a custom flyback transformer, calculating turns ratio, core selection, and parasitic considerations to optimize power transfer efficiency
- Performed component-level design and conducted full KiCad schematic simulation to validate performance prior to PCB fabrication
- Led hardware design reviews, served as primary point of contact with component vendors - directly communicating technical flyback transformer and power supply requirements to suppliers and evaluating component datasheets to make final selections

Wireless Charging Desk | Independent Project | Ongoing

- Designing a multi-device wireless charging system leveraging inductive resonant coupling via custom copper spiral coils - applicable to RF near-field energy transfer principles
- Developing PCB layout to interface power electronics with coil array, optimizing coil geometry for maximum coupling efficiency and minimal EMI
- Integrating a status display interface with embedded firmware for power regulation and device detection

LEADERSHIP/EXPERIENCE

Owner & Licensed Dealer | Used Car Dealership — San Jose, CA

- Sole proprietor and independently licensed dealer responsible for all aspects of dealership operations, including vehicle acquisition, sales, compliance, and end-to-end documentation management.

Owner & Technician | Motorcycle Resale Business — Facebook Marketplace

- Self-directed entrepreneur specializing in sourcing, restoring, and reselling motorcycles through Facebook Marketplace, managing all aspects of acquisition, mechanical refurbishment, and private sales.

Google Developer Student Club | Team Member | De Anza College

Jan 2022 – Aug 2022

- Collaborated in an agile team to build a cross-platform mobile app (Dart/Flutter) delivering budget tracking and event management features on schedule
- Facilitated team meetings and coordinated task distribution across engineering and design sub-teams

LANGUAGES

English (Fluent) | Russian (Fluent) | Georgian (Native) | Italian (Elementary) | French (Elementary)